

DENISON, IA, USA

WMO: 725477

Lat: 41.986N

Lon: 95.381W

Elev: 389

StdP: 96.74

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04917

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-17.7	-25.2	0.4	-19.6	-22.5	0.5	-17.1	13.2	-2.9	12.0	-4.6	2.4	330	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	32.9	24.5	32.2	24.3	30.3	23.3	26.8	31.2	25.6	29.9	24.6	28.9	5.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.3	21.4	29.8	24.1	19.9	28.4	22.9	18.5	27.4	86.5	31.4	81.3	29.7	76.5	29.2	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.1	8.8	-26.1	35.6	3.8	1.5	-28.8	36.6	-31.1	37.5	-33.2	38.4	-35.9	39.5		
(5)			-25.4	28.4	3.5	0.9	-27.9	29.0	-29.9	29.6	-31.8	30.1	-34.3	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-5.2	-3.7	3.1	9.9	15.9	21.8	23.8	22.1	18.6	10.9	3.5	-3.5		
	DBStd	11.65	6.72	6.97	6.82	5.57	4.93	3.68	3.40	3.06	4.78	5.63	6.00	6.47		
	HDD10.0	1844	470	383	231	68	7	0	0	0	2	57	207	419		
	HDD18.3	3618	728	616	473	258	108	11	3	5	55	239	444	677		
	CDD10.0	1784	0	1	17	65	189	354	427	374	259	85	13	1		
	CDD18.3	514	0	0	1	4	32	115	172	121	62	9	0	0		
	CDH23.3	4973	0	0	9	59	329	1161	1725	1026	574	88	2	0		
	CDH26.7	1714	0	0	1	11	100	418	663	316	183	22	0	0		
(14) Wind	WSAvg	3.8	4.0	4.1	4.6	5.0	4.3	3.6	2.9	2.7	3.2	3.9	4.0	3.9		
(15) Precipitation	PrecAvg	745	20	22	48	70	104	105	84	93	97	55	31	25		
	PrecMax	1009	57	89	123	214	240	315	206	200	280	121	92	87		
	PrecMin	383	0	1	4	17	33	20	9	7	17	0	0	2		
	PrecStd	171	15	20	33	45	46	54	49	52	68	37	26	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	17.0	24.1	28.7	32.9	34.1	35.1	34.0	32.9	29.2	22.6	12.9		
		MCWB	6.7	10.3	15.3	17.8	21.3	23.6	26.8	24.8	22.2	19.4	14.1	8.8		
	2%	DB	7.8	12.4	20.9	25.2	29.1	32.4	32.9	32.1	31.1	26.0	18.0	10.1		
		MCWB	4.7	8.2	13.9	15.6	20.1	23.3	26.1	24.8	22.3	17.6	12.2	6.7		
	5%	DB	5.2	8.6	17.0	22.5	27.2	31.0	32.2	30.0	28.8	22.7	15.2	7.4		
		MCWB	2.8	6.0	11.5	14.1	18.4	22.3	25.8	24.2	21.6	15.9	10.3	4.6		
	10%	DB	2.9	6.0	13.0	19.9	24.8	29.0	30.2	27.9	27.1	20.3	12.6	5.0		
		MCWB	1.3	3.4	8.4	13.1	17.5	21.6	24.7	22.8	20.2	14.4	8.3	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	12.6	19.2	19.9	23.5	26.6	28.6	27.2	25.3	21.9	16.0	11.2		
		MCDB	9.4	15.1	22.0	26.6	29.2	31.3	33.0	31.4	29.0	27.3	20.1	12.4		
	2%	WB	5.0	8.4	14.7	17.0	21.6	25.0	27.5	25.8	24.1	19.4	13.5	7.0		
		MCDB	7.5	12.0	17.8	22.5	27.3	29.9	31.8	30.0	28.2	23.5	17.4	8.8		
	5%	WB	3.1	5.6	11.7	15.4	20.0	23.9	26.5	24.7	22.9	17.5	11.0	4.8		
		MCDB	5.1	8.5	16.6	21.0	24.9	29.0	30.8	28.9	27.1	21.0	14.7	7.1		
	10%	WB	1.7	3.6	8.6	13.7	18.6	22.8	25.3	23.6	21.6	15.1	8.7	3.1		
		MCDB	3.1	5.9	14.0	18.6	23.4	27.7	29.4	27.6	25.5	19.1	12.6	4.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.4	11.2	11.9	11.5	11.3	10.5	10.7	12.5	11.8	10.8	8.8		
		MCDBR	11.8	12.5	16.4	16.3	14.5	12.4	11.1	11.7	13.5	14.8	14.7	11.9		
	5% WB	MCWBR	8.9	8.8	10.0	8.8	7.0	5.9	6.1	6.8	6.7	7.9	9.0	8.9		
		MCWBR	10.7	11.5	14.6	13.3	12.0	10.7	9.8	10.4	11.1	12.0	13.0	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.302	0.323	0.376	0.399	0.410	0.424	0.411	0.374	0.332	0.294	0.282		
	taud		2.422	2.378	2.403	2.275	2.281	2.306	2.254	2.296	2.359	2.464	2.496	2.443		
	Ebn at Noon		875	905	925	892	877	865	847	848	857	860	853	840		
	Edh at Noon		77	95	105	128	131	128	134	124	109	87	72	70		
(44) All-Sky Solar Radiation	RadAvg		1.93	2.95	3.79	4.81	5.71	6.33	6.51	5.54	4.55	3.12	2.15	1.61		
	RadStd		0.18	0.20	0.34	0.39	0.42	0.52	0.40	0.38	0.29	0.38	0.14	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page